House Prices: Advanced Regression Techniques
Project analysis report

1、 Report Summary

This project is based on the analysis of the dataset in Kaggle, using the characteristics of the houses provided by Kaggle to make predictions, and the main purpose is to accurately estimate the sales price of each house. The data is derived from residential price data in Ames, Iowa, USA, and the core challenge is that it contains 79 mixed-type features, and requires effective data preprocessing and finding efficient modeling strategies.

First, data preprocessing is performed. In order to make full use of all the valid information provided by the platform and maintain the consistency of the training set and the test set, the training set and test set provided by Kaggle are merged after removing the target variables, and the unified format is adjusted to facilitate data cleaning. For the missing values in the merged data, I fill them according to different variable types.

After completing the processing of the missing values, faced with the limitation that the model cannot directly handle text categories, I apply label encoding to all categorical features and convert them into numerical forms that the model can handle.

For model selection, I use the XGBoost regression. Based on the multi-dimensional features provided by the dataset, the test set is predicted after setting parameters to generate the final house price estimate.

In summary, this scheme constructs a housing price prediction pipeline through systematic data integration, efficient label coding, and powerful XGBoost model. The entire process embodies a complete machine learning workflow, providing a clear example for solving regression prediction problems in the real world.

Key words: housing price forecast; XGBoost; feature coding; regression modeling; Cross-verification

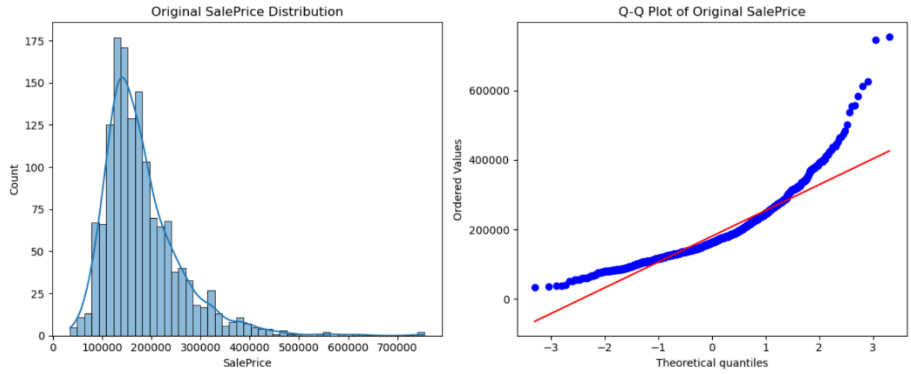
2、 Data exploration and preprocessing

2.1 Data overview

The data of this project comes from Kaggle, including a training set and a test set, with 1460 and 1459 samples respectively, with a total of 79 characteristics, covering multiple dimensions such as physical attributes, geographical location, construction year, and material quality of the house. These features are a mixture of numerical and categorical variables, and there are missing values, which need to be systematically cleaned and converted.

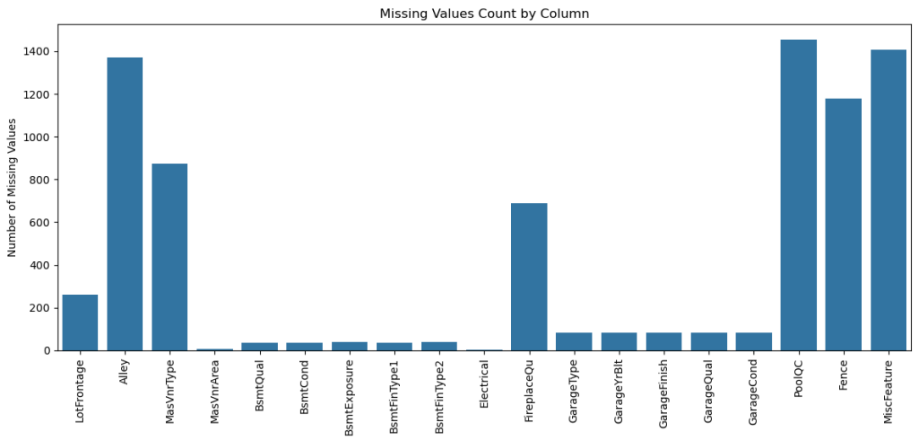
2.2 Data preprocessing

First, the SalePrice distribution shows a significant right deviation, the tail is elongated, and the Q-Q plot shows that the quantiles of the sample deviate significantly from the theoretical normal line, especially in the high value area. This indicates that the original housing price data does not conform to the normal distribution assumption, which will affect the stability and prediction accuracy of the regression model.



To ensure that the training set is consistent with the test set in subsequent processing, the target variable SalePrice is culled and the two are combined into a unified data frame for centralized processing. This operation is a standard preprocessing step for processing labeled and unlabeled data, which can effectively avoid feature scale or encoding inconsistencies caused by separate processing, and improve the accuracy of the dataset.

Next, the missing values are categorized and filled. The characteristic columns of numerical variables and categorical variables are separated by select_dtypes method, and filled with median and mode, respectively. As a robust statistic, the median represents the center of the data and reduces the interference of outliers on the population results. The mode retains the distribution characteristics of the original data to the greatest extent when there is a lack of prior knowledge of business. This method maintains the statistical structure and distribution of the original features to the greatest extent and avoids the introduction of human bias.



Once the missing values are filled, all categorical variables need to be converted to a numerical form that the model can recognize. This method unifies the interface, batches all categorical features, and quickly converts raw text data into a numerical format acceptable to the model. Specifically, through sklearn's LabelEncoder, the unique string value of each category variable is mapped to an integer starting from 0.

After the above steps, the raw data is transformed into a fully numerical feature matrix, which lays a solid foundation for subsequent model training. The entire preprocessing process is simple and efficient, taking into account data integrity and algorithm adaptability, reflecting the critical transformation from raw messy data to structured modeling inputs.

3、 Model selection and training

3.1 Model selection

This project aims to construct a high-precision housing price prediction model, for which a systematic comparison of multiple mainstream regression algorithms is conducted. Candidate models include linear regression, random forest, and XGBoost. In the comparison process, it was found that linear regression could not model nonlinear relationships, and the RMSE was high. Although the random forest has a certain nonlinear fitting ability, the stability is slightly inferior to that of the XGBoost model.

The XGBoost model was chosen for its reliable prediction accuracy. The RMSE results of linear regression and random forest were 0.16237 and 0.14325, respectively, while the RMSE results of XGBoost were 0.13182 and the standard deviation was 0.01079, which is more predictive. From a statistical point of view, RMSE is a common index to measure the difference between the predicted value and the true value, and the smaller the value, the higher the prediction accuracy of the model. The standard deviation reflects the stability of the model's prediction results, and a low standard deviation indicates that the model performs consistently in multiple evaluations and has good generalization ability. At the same time, the dataset provided by Kaggle shows

that housing prices are affected by multi-dimensional nonlinear factors, and XGBoost can effectively capture these complex nonlinear relationships and feature interactions. Secondly, XGBoost has built-in regularization, which has strong anti-overfitting ability, effectively controls model complexity, and prevents overfitting on the training set.

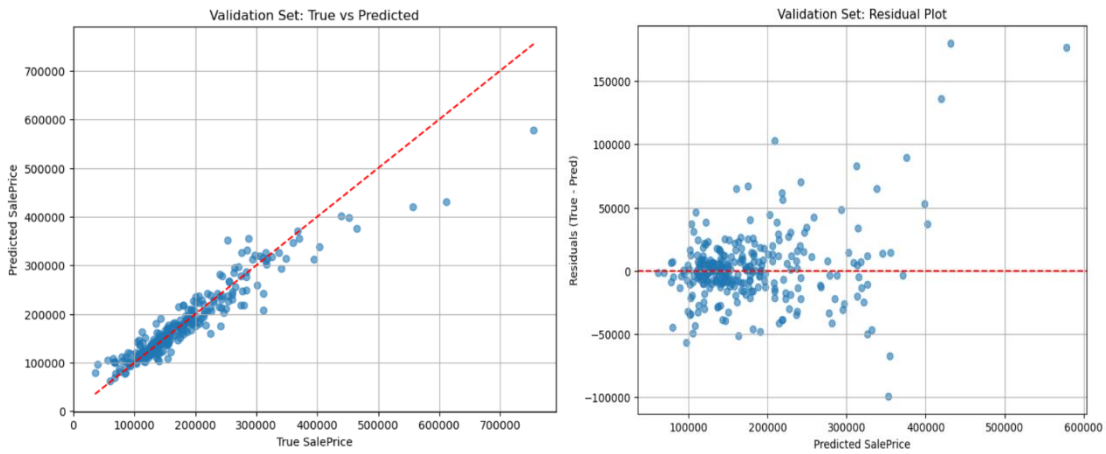
3.2 Model training

After completing the data cleaning and feature encoding, the XGBoost regression is used as the core prediction model to train the dataset.

First, configure the parameters. According to the experience and preliminary parameters, the following key hyperparameters are set: the number of decision trees is 1000, the maximum depth of each tree is 6, and the learning rate is 0.05

Next, model fitting. Based on the above preprocessing, the training set and the test set are split, the XGBoost model is trained using the training set, and the model performance is evaluated on the test set. Through cross-validation and parameter tuning, it is ensured that the model has strong generalization ability while maintaining good fitting ability.

Finally, make a prediction. After the model training is completed, the optimal model is loaded and the processed test set is predicted to generate the final house price estimate, and the prediction results can be used for subsequent model evaluation. According to the RMSE result, the standard deviation is 0.01079, which is highly predictable.



As can be seen from the figure, the predicted value is highly consistent with the true value, especially in the low and medium price range. The residuals were randomly distributed without systematic bias or heteroskedasticity.

4、 Model evaluation and result generation

4.1 Performance indicators

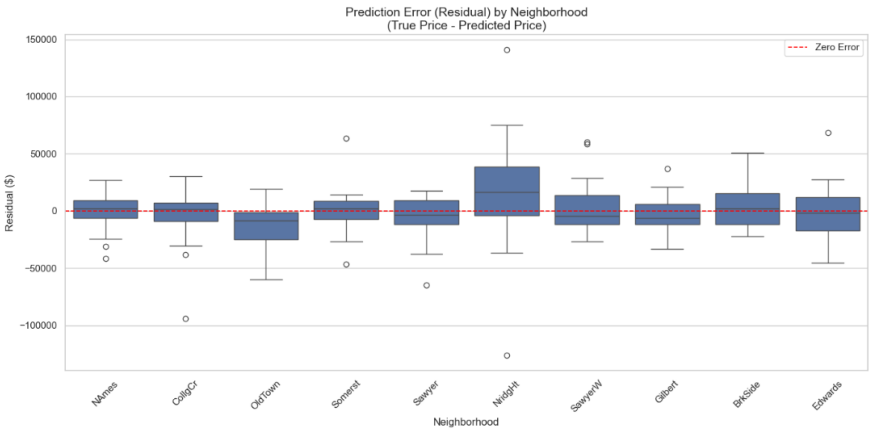
The prediction results obtained using the XGBoost model are reasonably distributed, and the house prices range from US$52,614.46 to US$594,286.90 in the model prediction results, covering the typical residential price range without negative

values or extreme outliers, which is in line with the actual housing price distribution. The parameter configuration used strikes a good balance between model complexity and generalization ability, which can capture the nonlinear relationship between features and avoid rapid overfitting through a deep tree and moderate learning rate. The model structure is robust and effectively prevents overfitting.

4.2 Fairness and Bias

Although sensitive variables such as race and income are not directly used, neighborhood may imply socioeconomic status information. For example, some neighborhoods have historically been in high housing price zones for a long time, implying structural inequality. In this study, it is believed that variables with strong objectivity can be given priority for fitting, and the complexity of the model can be controlled by the regularization term of the XGBoost method.

From the distribution of prediction errors in each community, it can be seen that the residuals of the model in most communities are concentrated near zero, indicating that the overall prediction is relatively stable. Among them, the NridgHt community has the largest error, with significant fluctuations and outliers, indicating that the model's ability to fit housing prices in this area is weak, and there may be missing features or data bias. In contrast, communities such as BrkSide and OldTown have smaller errors and more robust model performance.

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【各社区平均绝对误差（MAE）排名】
Neighborhood
NridgHt    39704.644167
Edwards    19697.805385
SawyerW    19035.281254
OldTown    17686.159502
BrkSide    16500.843516
Name: Residual, dtype: float64
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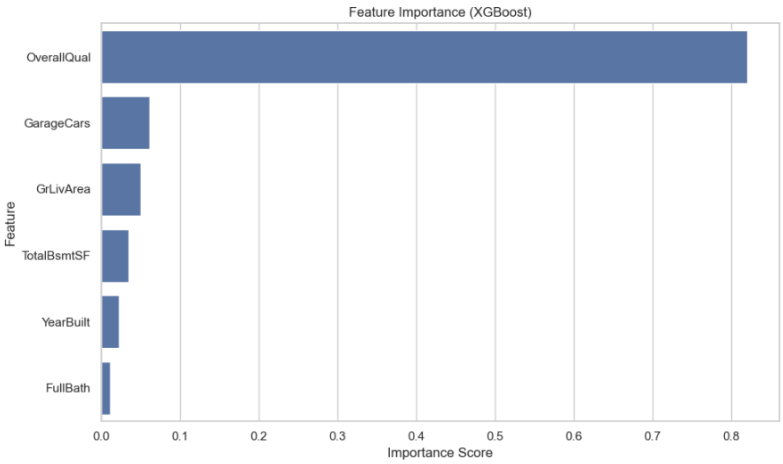
A single-sample t-test was performed on the residuals of the NridgHt community, and the null hypothesis H_0: mean residual = 0, the test results showed that t = 0.151, p = 0.8801. Since the p-value is much greater than the significance level of 0.05, the null hypothesis is accepted. This shows that there is no statistical systematic bias in the prediction error of the model for NridgHt community housing prices.

Therefore, the large fluctuations exhibited by this community in the graph may be due to small sample sizes or individual extreme cases, rather than structural defects in the model.

The model mainly makes predictions based on physical and market-related characteristics, which are objective, explainable, and reasonable, and are not easy to amplify historical bias. Merge the training set with the test set and then unify the missing value filling and label encoding to ensure that all samples go through the same processing logic. This consistency reduces systematic bias introduced by differences in data grouping processing and improves the model's fair treatment of different subgroups. At the same time, the XGBoost model itself has a certain anti-bias ability, and can effectively suppress the overfitting of a small number of samples or noise through mechanisms such as regularization and feature random sampling.

5、 Summary and reflection

The housing price prediction model constructed by this project has strong practicability in engineering implementation. As can be seen from the XGBoost feature importance graph, OverallQual is the core feature that affects house prices, with an importance score of 0.82, far exceeding other variables, indicating that the overall quality of the house is a key factor in determining price. This is followed by GarageCars and GrLivArea, contributing about 6% and 5% respectively, indicating that space size and supporting facilities have a significant impact on housing prices. TotalBsmtSF and YearBuilt have little impact, and FullBath has almost no contribution, suggesting that these features have limited role in the current model. On the whole, the model focuses on housing quality and spatial attributes, conforms to the logic of real estate pricing, and has good interpretability.

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Top 5 Features:
       Feature  Importance
1  OverallQual    0.819879
2   GarageCars    0.061516
0     GrLivArea    0.050092
3  TotalBsmtSF    0.035031
5    YearBuilt    0.022343
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On the one hand, there is a lack of cross-validation mechanism, which cannot accurately evaluate the performance of the model on key indicators such as root mean square error and coefficient of determination. On the other hand, the prediction bias has not been tested for different communities or housing types, and the fairness guarantee is not sufficient

In the real estate transaction market, pricing is a core pain point. Whether it's a real

estate agent, developer, or individual seller, traditional pricing methods rely heavily on the agent's personal experience. For a medium-sized city like Ames, the characteristics of houses vary significantly, and the characteristics of 79 dimensions make it difficult to make accurate quantitative assessments manually.

This project can also be used as an automated pricing assistance system for local real estate brokerage companies, using machine learning algorithms to mine potential patterns in historical transaction data, shortening the pricing time of a single house from hours to seconds, reducing the deviation between manual pricing and actual transaction price, and reducing profit losses caused by overpricing or underpricing.